

Ex01. Introduction to Signals and systems

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1.1. Develop 8 Exponential Signals As a Graph

This assignment involves creating eight graphs representing exponential signals. These signals are discrete-time and defined over the range of integers n from -100 to 100. The eight graphs are generated based on combinations of three factors: whether the real or imaginary part of the complex signal is plotted, the presence of noise, and the sign of the damping factor (or real part of the complex frequency) σ .

```
n = -100:100;
T0 = 0.5;
Ts = 0.01;
sigma_values = [+1, -1];
w0 = 2*pi/T0;

figure('Position', [100, 100, 1000, 600]);

sigma = sigma_values(1);
x = exp((sigma + j * w0)* Ts * n);

% Calculate the real and imaginary parts of the signal
realPart = real(x);
imagPart = imag(x);

% Define noise
noise = 1.0 * rand(size(x));
realPart_noise = noise + realPart;
imagPart_noise = noise + imagPart;

subplot(2,4,1);
p1 = plot(n, realPart);
title(strcat("real(x[n]), \sigma = ", num2str(sigma)));
xlabel("n");
ylabel("Amplitude");
axis([-100 100 -4 4]);
set(p1, 'Color', 'b');
set(p1, 'Marker', '^');
grid on;

subplot(2,4,2);
p2 = plot(n, imagPart);
title(strcat("imag(x[n]), \sigma = ", num2str(sigma)));
xlabel("n");
ylabel("Amplitude");
axis([-100 100 -4 4]);
set(p2, 'Color', '[0.3 0.8 1.0]');
set(p2, 'Marker', '+');
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grid on;

subplot(2,4,3);
p3 = plot(n, realPart_noise);
title(strcat("real(x[n]) + noise, \sigma = ", num2str(sigma)));
xlabel("n");
ylabel("Amplitude");
axis([-100 100 -4 4]);
set(p3, 'Color', 'g');
set(p3, 'Marker', 'o');
grid on;

subplot(2,4,4);
p4 = plot(n, imagPart_noise);
title(strcat("imag(x[n]) + noise, \sigma = ", num2str(sigma)));
xlabel("n");
ylabel("Amplitude");
axis([-100 100 -4 4]);
set(p4, 'Color', 'k');
set(p4, 'Marker', 'square');
grid on;

sigma = sigma_values(2);
x = exp((sigma + j * w0)* Ts * n);

% Calculate the real and imaginary parts of the signal
realPart = real(x);
imagPart = imag(x);

% Define noise
noise = 1.0 * rand(size(x));
realPart_noise = noise + realPart;
imagPart_noise = noise + imagPart;

subplot(2,4,5);
p1 = plot(n, realPart);
title(strcat("real(x[n]), \sigma = ", num2str(sigma)));
xlabel("n");
ylabel("Amplitude");
axis([-100 100 -4 4]);
set(p1, 'Color', '[1.0 0.4 0.7]');
set(p1, 'Marker', '^');
grid on;

subplot(2,4,6);
p2 = plot(n, imagPart);
title(strcat("imag(x[n]), \sigma = ", num2str(sigma)));
xlabel("n");
ylabel("Amplitude");

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axis([-100 100 -4 4]);
set(p2, 'Color', 'red');
set(p2, 'Marker', '+');
grid on;

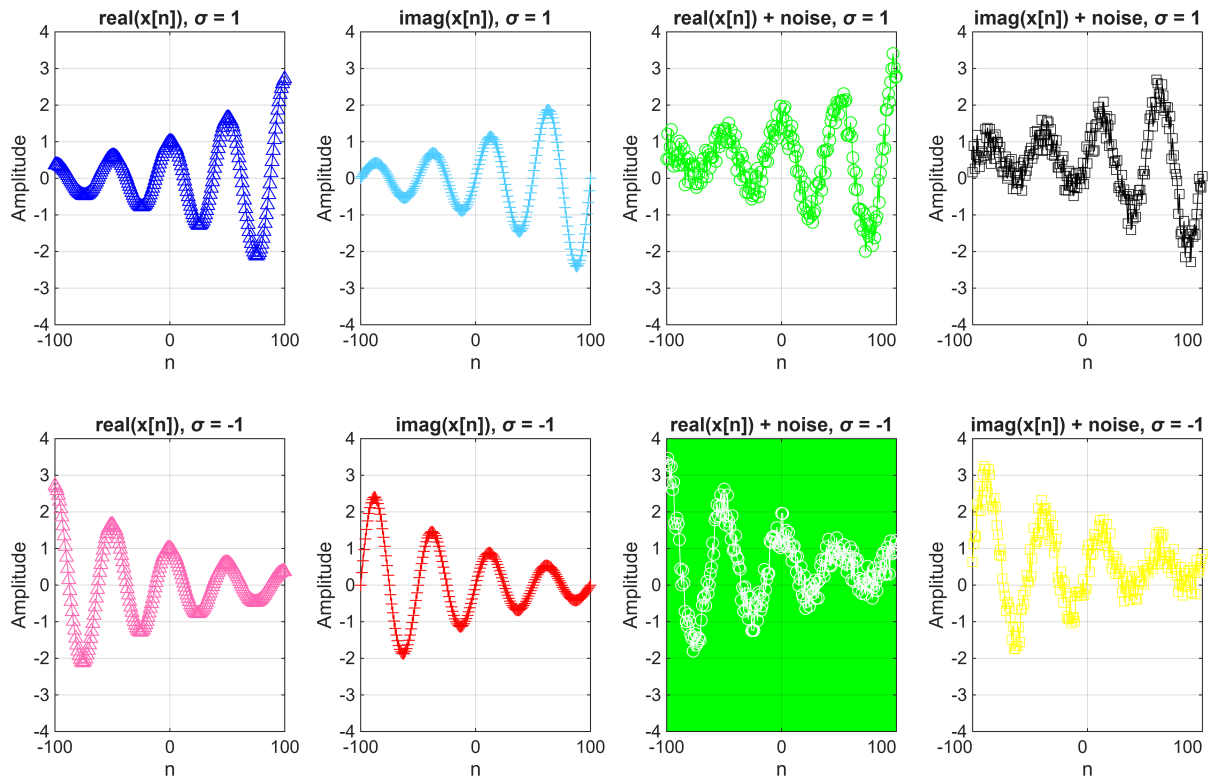
subplot(2,4,7);
p3 = plot(n, realPart_noise);
title(strcat("real(x[n]) + noise, \sigma = ", num2str(sigma)));
xlabel("n");
ylabel("Amplitude");
axis([-100 100 -4 4]);
set(p3, 'Color', 'w');
set(p3, 'Marker', 'o');
ax = gca;
set(ax, 'Color', 'g');
grid on;

subplot(2,4,8);
p4 = plot(n, imagPart_noise);
title(strcat("imag(x[n]) + noise, \sigma = ", num2str(sigma)));
xlabel("n");
ylabel("Amplitude");
axis([-100 100 -4 4]);
set(p4, 'Color', 'yellow');
set(p4, 'Marker', 'square');
grid on;

%% Below is the 8 graph required in the assignemnt.
sgtitle('x[n] = e^{(\sigma + j\omega_0)T_s n} Analysis');

```

$$x[n] = e^{(\sigma + j\omega_0)T_s n} \text{ Analysis}$$



1.2. Math Problem

Below is the snapshot of my answer in fill-blank math problem.

$$\begin{array}{c}
 e^{j\omega_0 n T_s} \times e^{j\omega_0 n_0 T_s} \\
 \hline
 \frac{m}{N_0} \quad \frac{T_s}{T_0}
 \end{array}
 \quad
 \frac{e^{j\omega_0 n T_s}}{2\pi m}$$